Investigation Report

INV-002: Response-surface models below the model-adequacy criteria

Quality Assurance (QA)

2026-09-28

Table of contents

Approvals 4
Revision history 5
Abbreviations 5

Summary **6**

1 Event description **8**

2 Scope **10**

3 Finding **12**

4 Root cause **15**

5 Remedy and decision rule **17**

6 Sample-size calculation **22**
6.1 Method 22
6.2 D-optimal augmentation 23
6.3 Results 24
6.4 Power of the effects 27

7 Augmentation designs **28**
7.1 Production bioreactor 28
7.2 Anion exchange 29

8 Model selection (corrective action CA3) **31**
8.1 Procedure 31
8.2 Why a log scale, and what it changes 32
8.3 How the model is searched 34
8.4 Verification of the platform worst cases 35
8.5 What model selection changed 36
8.5.1 The augmentations 39
8.5.2 Outputs with no adequate model 41
8.5.3 An output that fits exactly 42
8.5.4 Outputs with no detected effect 43

9 Impact assessment **44**
9.1 Acceptance criteria 46
9.2 Conclusions over the NOR 49
9.3 Other documents 50

10 Corrective and preventive actions **51**

11 Effectiveness check **52**

12 Conclusion and closure **53**

References **54**

Field	Value
Document ID	INV-002
Document class	Investigation Report
Title	Investigation Report: A-Mab Drug Substance
Product	A-Mab
Version	1.0
Effective date	2026-09-28
Supersedes	—
Status	Approved (synthetic)
Document owner	Quality Assurance (QA)
Sponsor	Novacyte Biologics
Development site	Cambridge, MA
Commercial drug substance site	Grafton, WI

> **SYNTHETIC DOCUMENT — NOT A REAL RECORD.** Illustrative content generated from a seeded model of the *A-Mab* case study (CMC Biotech Working Group, 2009). Novacyte Biologics and all sites, SOP/AMV numbers and signatories are fictional. For NLP corpus development only; not for any regulated use.

Approvals

Role	Function	Name (synthetic)	Signature	Date
Author	Quality Assurance	—	—	—
Reviewer	Statistics	—	—	—
Reviewer	Manufacturing Science & Technology	—	—	—
Reviewer	Analytical Development	—	—	—
Approver	Quality Assurance	—	—	—

Synthetic record. Signatures are not applied.

Revision history

Version	Effective date	Description of change
1.0	2026-09-28	Initial issue

Abbreviations

Abbreviation	Meaning
AIC	Akaike information criterion
AICc	Akaike information criterion, corrected for small samples
AMV	analytical method validation
BLA	Biologics License Application
CA	corrective action
CE-SDS	capillary electrophoresis with sodium dodecyl sulfate
CV	column volumes
DoE	design of experiments
ELISA	enzyme-linked immunosorbent assay
Equil	equilibration
GMP	good manufacturing practice
HMW	high molecular weight
INV	investigation (in document identifiers such as INV-002)
LMW	low molecular weight
LRF	log reduction factor
MSAT	Manufacturing Science and Technology
MVM	minute virus of mice
NOR	normal operating range
PA	preventive action
PCMP	process characterization master plan
PCMR	process characterization master report
pCO2	CO2 partial pressure
PCR	process characterization report
PD	plan discrepancy (in identifiers such as PD-003-01)
PRESS	prediction error sum of squares
QA	Quality Assurance
RSM	response surface methodology (in study codes such as S-VF-RSM)
R^2	coefficient of determination
SD	standard deviation
SOP	standard operating procedure
UF/DF	ultrafiltration and diafiltration
XMuLV	xenotropic murine leukemia virus

Summary

The Version 2.0 re-analysis of the A-Mab characterization campaign tested every response-surface model that carries an acceptance criterion at its step against the model-adequacy criteria of the process characterization master plan (PCMP-001 §7.2). The re-analysis applied three of the four criteria as tests, which require a significant model F-test ($\alpha = 0.05$), no significant lack of fit ($\alpha = 0.05$) and R^2 minus predicted R^2 of at most 0.20 (Section 3). Of the 22 full quadratic models in scope, 14 did not meet the limit on overfitting, which allows predicted R^2 to fall short of R^2 by at most 0.20. Besides these 14 models, the full quadratic model of acidic variants at low-pH viral inactivation reproduced every run exactly, so none of the tests could be applied to it. With Protein A response-surface run 23 at its design elution buffer pH, as Version 1.0 fitted it, no model showed significant lack of fit. At the pH recorded for Protein A run 23, the full quadratic model of pool host cell protein failed the lack-of-fit test. QA opened INV-002 on 2026-08-31 and classified it as minor, on the grounds recorded at opening (no GMP batch was made or dispositioned on the analysis, no product quality impact, and the finding concerns the statistical analysis of development data, not their integrity).

The root cause is that the plans prescribed the full quadratic model as the predictive model of each design, with no model selection. The full quadratic model keeps every term whatever the data show, and the planned designs are small for a model with that many terms. The Version 1.0 plans also set no numeric limit on how far predicted R^2 may fall below R^2, against which the size of a design could have been checked.

A rule for the remedy of each failing model was approved on 2026-09-02, before any new run was made (Section 5). A failing model of an attribute of the drug substance specification, on which the refit of the planned runs keeps an effect, is resolved by further runs (corrective action CA1), unless the target probability of success cannot be reached within the size of the planned design. Every other failing model is refitted by hierarchical backward elimination (corrective action CA2). Viral clearance is demonstrated by spiking studies against a cumulative claim, not by testing the drug substance, so its models are refitted and no virus is spiked.

The remedy rule gives 3 models an augmentation, in 2 steps. The production bioreactor design was augmented by 16 runs for high mannose and acidic variants, in study S-BRX-AUG. The anion exchange design was augmented by 12 runs for host cell protein, in study S-AEX-AUG. Each augmentation is the smallest D-optimal set of runs that gives a probability of success of at least 0.90. The probability of success is the probability, at the effects estimated from the planned runs, that the planned model, the full quadratic with a block term, meets the lack-of-fit test and the limit on overfitting on all runs (Section 6). On all runs that model meets both tests for acidic variants

and host cell protein, and it does not meet the limit on overfitting for high mannose (Section 8.5.1). The other 11 failing models were refitted.

Model selection was adopted on 2026-09-15 as corrective action CA3, for every response-surface output (Section 8). Each output is modelled on the log or the raw scale, as its Box-Cox profile decides, by the submodel of the full quadratic with the lowest AICc, and that model is judged against the three tests. Model selection replaced the refits. Once selected, the three models that the augmentations were made for meet the three tests on the planned runs alone, so the augmentations were not needed to pass them. The augmentation runs removed from two models of the production bioreactor, those of high mannose and aggregates, terms that the planned runs had suggested. Their models selected on the planned runs had predicted only 13 and 11 of the 16 new runs within the 95 % prediction interval, fewer than a one-sided binomial test at the 0.05 level allows. The reports use the models of all runs. A check of a selected model against the three tests is optimistic, because the selection used the same runs. After selection, 4 outputs have no adequate model (acidic variants at low-pH viral inactivation, fragments at low-pH viral inactivation, fragments at cation exchange and XMuLV clearance at virus filtration), and they are judged on their measured results.

Model selection was adopted before the investigation into acceptance criteria set after the data were seen (INV-001) closed on 2026-09-18, so the intermediate acceptance criteria fixed at that closure rest on the selected models. The verification of the platform worst cases widened 3 of these criteria, for aggregates at the production bioreactor, host cell protein in the Protein A pool and host cell protein at cation exchange, by replacing a platform worst case with the one-sided 95 % prediction bound of a single A-Mab batch. The criterion for aggregates at the production bioreactor was also tightened by another downstream step, and it is narrower than its a priori value. Against the criteria that the same verification gives on the full quadratic models, model selection made 4 criteria wider, by 4 %, 4 %, 7 % and 7 %, and kept the upper limit of acidic variants at the production bioreactor at its a priori value, which the full quadratic models would have widened (Section 9.1). Against the criteria without model selection, no output changes whether it meets its criterion at every combination of NOR low, set-point and NOR high, and one proven acceptable range moves. Against the criteria without model selection, however, the scale-down model would not be qualified for host cell protein in the Protein A pool and at cation exchange. The augmentation runs moved no criterion.

This report gives the models of the augmented steps on all runs and how well the models selected on the planned runs predict the new runs. This report also gives two prediction limits from the models of the augmented steps on all runs. One is the lower prediction limit of acidic variants at the production bioreactor at the least favourable combination of the NOR (Section 9.2). The other is the one-sided 95 % prediction bound of a single batch for host cell protein at anion exchange (Section 9.1). The reports of the two augmented steps give the measured results of the new runs, the re-evaluation of every model and the ranges.

1 Event description

INV-001, the investigation into acceptance criteria set after the data were seen, was opened on 2026-08-03. It followed the QA review of the process characterization master report (PCMR-001) Version 1.0 for BLA readiness, which found that the acceptance criteria of the proven acceptable range analysis had been set after the data were seen. Its corrective actions included the reissue of every characterization report at Version 2.0, re-analysed against criteria set a priori. Version 2.0 of the process characterization master plan (PCMP-001) also states campaign thresholds for model adequacy (PCMP-001 §7.2). Version 1.0 of the master plan accepted a response-surface model when its terms were significant, its lack of fit was not significant against pure error and its predicted R^2 was consistent with its adjusted R^2. Version 1.0 of the master plan left the numeric thresholds to the individual study. The Version 1.0 plans set no numeric limit on how far predicted R^2 may fall below R^2, against which the size of a design could have been checked. The Version 1.0 reports fitted the full quadratic model to every response-surface design and applied the lack-of-fit test.

The Version 2.0 re-analysis tested every response-surface model with an acceptance criterion at its step against three of the four model-adequacy criteria of the master plan. These three tests require a significant model F-test ($\alpha = 0.05$), no significant lack of fit ($\alpha = 0.05$) and R^2 minus predicted R^2 of at most 0.20 (Section 3). The re-analysis found 14 of the 22 full quadratic models above the limit on overfitting. QA opened INV-002 on 2026-08-31, while INV-001 was still open. The recorded trigger is the re-analysis itself.

As an immediate action, no proven acceptable range, design space or judgement across the normal operating range (NOR) was set from a model that failed one of the three tests of the model-adequacy criteria while the investigation was open. The Version 1.0 documents, whose ranges came from the full quadratic models, were already on hold for BLA use under INV-001.

The remedy rule, which assigns a remedy to each failing model, was approved within this investigation on 2026-09-02, together with the sample sizes and the augmentation designs. The first augmentation run started on 2026-09-03. The root cause was recorded on 2026-09-15, and model selection was adopted on the same date as corrective action CA3. Table 1.1 gives the dates of the investigation.

Table 1.1: Chronology of INV-002 and of the augmentation studies.

Date	Event
2026-05-15	Last planned response-surface study ended, the virus filtration response-surface study (S-VF-RSM)
2026-08-03	INV-001 opened on the acceptance criteria of the proven acceptable range analysis
2026-08-31	INV-002 opened on the response-surface models below the model-adequacy criteria
2026-09-02	Rule for the remedy of each failing model approved, with the sample sizes and the augmentation designs
2026-09-03	Production bioreactor augmentation study started (S-BRX-AUG)
2026-09-10	Anion exchange augmentation study started (S-AEX-AUG)
2026-09-15	Root cause recorded, and model selection adopted as corrective action CA3
2026-09-18	INV-001 closed, with the intermediate acceptance criteria fixed on the selected models of the planned designs
2026-09-24	Both augmentation studies ended (S-BRX-AUG, S-AEX-AUG)
2026-09-28	INV-002 closed

2 Scope

The investigation covers every response-surface model that carries an acceptance criterion at its step. These are 22 models in the 6 steps with a designed experiment (Table 2.1). The finding is that 14 of the full quadratic models, fitted to the runs of the planned face-centred design as the process characterization master plan prescribes (PCMP-001 §6.2), exceed the limit of 0.20 on R^2 minus predicted R^2. Model selection, corrective action CA3, later analysed the same outputs (Section 8).

Table 2.1: Response-surface models in the scope of INV-002, by step.

Step	Runs	Terms	Attributes with a criterion	Of these, full quadratic not adequate
Production bioreactor	28	15	Afucosylation, galactosylation, high mannose, acidic variants, aggregates	High mannose, acidic variants
Protein A	28	15	Host cell protein, leached Protein A, fragments	Leached Protein A, fragments
Low-pH viral inactivation	18	10	Aggregates, XMuLV clearance, acidic variants, fragments	XMuLV clearance, acidic variants[a)], fragments
Cation exchange	28	15	Aggregates, host cell protein, fragments	Fragments
Anion exchange	28	15	Host cell protein, XMuLV clearance, MVM clearance, fragments	Host cell protein, XMuLV clearance, MVM clearance, fragments
Virus filtration	12	6	MVM clearance, XMuLV clearance, fragments	MVM clearance, XMuLV clearance, fragments

[a)] Fits exactly, so no test of the model-adequacy criteria can be applied, and the model is not adequate.
Runs: the runs of the face-centred design approved in the plan of the step. Terms: the terms of the full quadratic model, intercept included. Not adequate: the full quadratic model fails at least one of the three tests of PCMP-001 §7.2 that this investigation applies, a significant model F-test ($\alpha = 0.05$), no significant lack of fit ($\alpha = 0.05$) and R^2 minus predicted R^2 of at most 0.20, on the planned runs, with Protein A response-surface run 23 at its design elution buffer pH.

The following are outside the scope:

1. The screening models. They identify effects and set no range (PCMP-001 §6.2).
2. The rule that derives the intermediate acceptance criteria. The investigation into acceptance criteria set after the data were seen (INV-001) set that rule, and this investigation does not change it. Section 9.1 reports how model selection entered the verification behind the criteria, and which criteria it moved.
3. Responses with no criterion at their step, such as step yield. The reports select their models in the same way, and no conclusion depends on them.

4. The univariate studies, and the two steps without a designed experiment (harvest and UF/DF).
5. The measured results of the augmentation runs and the ranges that follow from them. The reports of the augmented steps give them.

Four terms are used throughout:

1. A conclusion over the NOR is the judgement whether an output meets its criterion at every combination of NOR low, set-point and NOR high of the response-surface factors.
2. The NOR as characterized is the NOR against which the studies were planned and run.
3. The NOR in force is the NOR that a Version 2.0 report sets, where it moved one.
4. The single-batch bound is the one-sided 95 % prediction bound of a single batch.

3 Finding

The process characterization master plan sets four model-adequacy criteria for a response-surface model (PCMP-001 §7.2):

1. The model F-test is significant at $\alpha = 0.05$.
2. Lack of fit is not significant against pure error at $\alpha = 0.05$.
3. Predicted R^2 falls short of R^2 by at most 0.20. This is the limit on overfitting.
4. The residuals show no pattern against predicted values or run order, and no single run dominates the fit.

The lack-of-fit test splits the residual variation into pure error, from the replicated centre points, and lack of fit, which is the rest. The prediction error sum of squares (PRESS) adds up the squared error of each run when it is predicted from a fit to all the other runs, and predicted R^2 is one minus PRESS divided by the total sum of squares (Montgomery 2017). For a model that fails, the master plan states the remedy in these words. “A model that fails the lack-of-fit test or the limit on overfitting is either refitted with a justified change of terms or transformation, or it is not used to set a range.”

This investigation applies the first three criteria as tests, because they decide whether a model of a design of this size can be adequate at all. The fourth criterion is judged by each step report for its own models. The Protein A report (PCR-005) Version 2.0, for example, records as discrepancy PD-005-02 that some of its conclusions rest on models that do not meet the fourth criterion. In this document a model is called adequate when it meets the three tests. The re-analysis applied the three tests to the full quadratic model of every output in scope, on the planned runs. The re-analysis fitted Protein A response-surface run 23 at its design elution buffer pH, as Version 1.0 had fitted it.

Table 3.1 gives the full quadratic model of each of the 22 outputs in scope, with its verdict. Each step with a response-surface design has at least one model that fails. The 14 failing models all fail the limit on overfitting, with R^2 minus predicted R^2 from 0.26 to 2.14 against a limit of 0.20. With Protein A run 23 at its design pH, no model fails the lack-of-fit test, and the lowest lack-of-fit p-value among the failing models is 0.096. The lack-of-fit test therefore gives no sign of systematic error in these full quadratic models. With 4 centre points, however, pure error has 3 degrees of freedom, so the test has little power to find such an error. Of the failing models, 5 also fail the model F-test, with p-values from 0.280 to 0.936. They are the 4 models of attributes with no detected effect and fragments in the Protein A pool. All of them fail the limit on overfitting as well, and that failure decides the remedy. Section 4 gives the cause.

Table 3.1: Full quadratic models of the planned runs against three of the model-adequacy criteria of PCMP-001.

Step	Attribute	Runs	Terms	R^2	Pred. R^2	Difference	p-value		Verdict
							Lack of fit	F-test	
Production bioreactor	Afucosylation (%)	28	15	0.96	0.79	0.17	0.182	<0.001	adequate
	Galactosylation (%)	28	15	0.97	0.92	0.06	0.992	<0.001	adequate
	High mannose (%)	28	15	0.91	0.65	0.26	0.748	<0.001	not adequate
	Acidic variants (%)	28	15	0.94	0.60	0.34	0.147	<0.001	not adequate
	Aggregates, HMW (%)	28	15	0.97	0.83	0.13	0.568	<0.001	adequate
Protein A	Host cell protein (ng/mg)	28	15	0.99	0.93	0.06	0.058	<0.001	adequate
	Leached Protein A (ppm)	28	15	0.57	−0.71	1.28	0.953	0.365	not adequate
	Fragments, LMW (%)	28	15	0.60	−1.32	1.92	0.644	0.280	not adequate
Low-pH viral inactivation	Aggregates, HMW (%)	18	10	1.00	0.96	0.04	0.535	<0.001	adequate
	XMuLV clearance (log10)	18	10	0.90	0.58	0.32	0.580	0.004	not adequate
	Acidic variants (%)	18	10	1.00	1.00	0.00	n.t.	n.t.	not testable: exact fit[a)]
	Fragments, LMW (%)	18	10	0.85	0.12	0.73	0.505	0.016	not adequate
Cation exchange	Aggregates, HMW (%)	28	15	0.98	0.82	0.15	0.158	<0.001	adequate
	Host cell protein (ng/mg)	28	15	0.97	0.83	0.14	0.679	<0.001	adequate
	Fragments, LMW (%)	28	15	0.76	−0.40	1.16	0.785	0.031	not adequate
Anion exchange	Host cell protein (ng/mg)	28	15	0.93	0.66	0.27	0.499	<0.001	not adequate
	XMuLV clearance (log10)	28	15	0.92	0.62	0.29	0.911	<0.001	not adequate
	MVM clearance (log10)	28	15	0.90	0.39	0.50	0.619	<0.001	not adequate
	Fragments, LMW (%)	28	15	0.32	−1.83	2.14	0.855	0.936	not adequate
Virus filtration	MVM clearance (log10)	12	6	0.95	0.53	0.42	0.096	<0.001	not adequate
	XMuLV clearance (log10)	12	6	0.51	−1.14	1.65	0.655	0.398	not adequate
	Fragments, LMW (%)	12	6	0.27	−1.74	2.01	0.785	0.811	not adequate

[a)] The model reproduces every run. With no residual there is no pure error for the lack-of-fit test and no residual mean square for the model F-test (n.t., not testable).

Full quadratic model on the planned runs, as Version 1.0 fitted it, with Protein A response-surface run 23 at its design elution buffer pH. Terms include the intercept. Pred. R^2: predicted R^2. Difference: R^2 minus predicted R^2, calculated before rounding. Verdict: adequate when the model meets the three tests of PCMP-001 §7.2 that this investigation applies, a significant model F-test ($\alpha = 0.05$), no significant lack of fit ($\alpha = 0.05$) and R^2 minus predicted R^2 of at most 0.20.

The Version 2.0 analysis puts Protein A response-surface run 23 at the elution buffer

pH recorded under study event DEV-005-01, 3.38 instead of the design pH of 3.55 (discrepancy PD-005-03). With run 23 at its recorded pH, the full quadratic model of pool host cell protein fails the lack-of-fit test ($p = 0.024$), while it meets the model F-test and the limit on overfitting. With run 23 at either pH, the same 14 models fail the limit on overfitting. The selected model of pool host cell protein, fitted with run 23 at its recorded pH, is the one Table 8.1 reports, and it is adequate. Model selection, corrective action CA3, covers every output (Section 8), so the failing full quadratic model of pool host cell protein needs no separate remedy.

The failing models fall into two groups, which differ in what a remedy can achieve:

1. Models with an effect that the design is too small for (10 models). The refit of the planned runs by backward elimination (Section 5) keeps at least one term that is significant at $\alpha = 0.05$. The full models of this group reach an R^2 of 0.60 to 0.95.
2. Models of attributes with no detected effect within the range studied (4 models). The refit of the planned runs keeps no term besides the intercept. The model F-test of the full model is not significant for any of them, and predicted R^2 is below zero, down to -1.83 for the fragments model of anion exchange.

Fragments appear in both groups. The refit keeps an effect on fragments at Protein A, low-pH viral inactivation and cation exchange, and none at anion exchange and virus filtration. Table 5.1 gives the terms that each refit keeps.

Of the other 8 full quadratic models in Table 3.1, 7 meet all three tests and need no remedy. The re-analysis counted the last one, of acidic variants at low-pH viral inactivation, among them as well, because none of its tests failed. That model reproduces every run exactly. With no residual there is no pure error for the lack-of-fit test and no residual mean square for the model F-test, so neither test can be applied, and an exact fit is not adequate. The remedy rule gave it no remedy, and model selection judges the output on its measured results (Section 8.5.3). Like every other output, the 7 models that meet the tests are analysed by model selection under corrective action CA3 (Section 8).

4 Root cause

The primary cause is that the plans prescribed the full quadratic model as the predictive model of every response-surface design, with no model selection. The full quadratic model keeps every main effect, two-factor interaction and squared term, whatever the data show. The planned designs are small for a model with that many terms. With four factors, the full quadratic model has 15 terms, including the intercept, and the planned face-centred design has 28 runs. Low-pH viral inactivation has three factors, 10 terms and 18 runs. Virus filtration has two factors, 6 terms and 12 runs.

Predicted R^2 comes from predicting each run from a fit that leaves that run out. The weight a run has in its own fitted value is its leverage. The average leverage equals the number of terms divided by the number of runs, and in these designs it lies between 0.50 and 0.56. PRESS divides the residual of each run by one minus its leverage. Leverage differs from run to run, from 0.12 to 0.79 in these designs, so the enlargement at the average leverage understates the effect. In expectation, PRESS is the residual sum of squares times the sum of $1/(1 - \text{leverage})$ over the runs, divided by the residual degrees of freedom. That factor is 5.2 to 7.0 for these designs. R^2 minus predicted R^2 equals PRESS minus the residual sum of squares, divided by the total sum of squares. It is therefore expected to be 4.2 to 6.0 times the share of the variation that the model leaves unexplained, one minus R^2. The limit of 0.20 is then met, in expectation, only where R^2 is above 0.95 to 0.97. Every full quadratic model that met all three tests (a significant model F-test ($\alpha = 0.05$), no significant lack of fit ($\alpha = 0.05$) and R^2 minus predicted R^2 of at most 0.20) has an R^2 of at least 0.96, and every model that failed the limit on overfitting has an R^2 of at most 0.95.

The sample-size simulation of Section 6 quantifies this gap for the models that the remedy rule augments. With the planned runs alone, the probability that the full quadratic model meets the lack-of-fit test and the limit on overfitting is 0.251 for high mannose, 0.444 for acidic variants and 0.493 for host cell protein at anion exchange. These probabilities are computed at the effects that the same planned runs estimated.

Attributes with no detected effect within the range studied fail for a second reason. For such an attribute the full quadratic model fits only random variation, and each of its terms takes up part of it. R^2 stays between 0.27 and 0.57 while predicted R^2 is below zero. With no effect, R^2 minus predicted R^2 depends on the number of terms against the number of runs, not on the size of the noise, so it shrinks only as the runs outgrow the terms. The fragment models of Protein A and cation exchange behave in the same way with a small effect. Their probability of success is at most 0.005 at every size computed, up to the size of the planned design (Table 6.1).

The contributing cause is that no plan checked the size of its design against the model it was to fit. Version 1.0 of the process characterization master plan (PCMP-001) left the model-adequacy threshold to the individual study, and the Version 1.0 plans set none. The limit of 0.20 on R^2 minus predicted R^2 came with Version 2.0 of the master plan. The last planned response-surface study ended on 2026-05-15, before the investigation into acceptance criteria set after the data were seen (INV-001) was opened.

The root cause was recorded on 2026-09-15. Corrective action CA3, model selection for every response-surface output, answers it (Section 8). The remedy rule of Section 5 and the augmentations it gave had been decided before that date, under the full quadratic model that the plans prescribed.

5 Remedy and decision rule

This investigation judges a model by the three tests of the model-adequacy criteria of the process characterization master plan (PCMP-001 §7.2), which require a significant model F-test ($\alpha = 0.05$), no significant lack of fit ($\alpha = 0.05$) and R^2 minus predicted R^2 of at most 0.20. For a model that fails the lack-of-fit test or the limit on overfitting, the master plan allows a refit with a justified change of terms or transformation, and otherwise the model is not used to set a range (Section 3). The master plan names no augmentation. Augmentation is this investigation's rule, for a design too small for the model a decision needs. The A-Mab case study names the same remedy for a design that cannot resolve the terms a decision needs, which is to augment it (CMC Biotech Working Group 2009).

QA approved the remedy rule on 2026-09-02, before any new run was made, and it was applied to every failing model in the same way:

1. A failing model of an attribute of the drug substance specification, on which the refit of the planned runs keeps an effect, is resolved by further runs. The number of runs is the smallest that gives a probability of success of at least 0.90 that the planned model, the full quadratic with a block term, meets the lack-of-fit test and the limit on overfitting on all runs (Section 6).
2. Viral clearance is not tested on the drug substance. It is a process requirement, demonstrated by spiking studies at worst-case conditions against a cumulative claim (International Council for Harmonisation 2023). Its models are refitted, and no augmentation spikes virus.
3. No augmentation is larger than the planned design. Where the target probability of success is not reached within that size, the model is refitted.
4. Every other failing model is refitted.
5. A response with no adequate model, full or refitted, is judged on its measured results. No model sets a range for it.

For an attribute of the specification with an effect, a refit on a design that is too small can drop a real interaction or curvature, and the ranges of the step would then rest on a model that misses it. Further runs make the estimate of every term of the full model more precise. For an attribute with no detected effect, the refit to the mean describes the results, and further runs would only raise the power to detect a small effect. The target of 0.90 is the probability of success customary for a confirmatory study, 0.80 to 0.90, taken at its upper end because a further failure would leave an attribute with a criterion without an adequate model. The cap on the size of an augmentation, the size of the planned design, keeps a remedy from becoming a new study.

The remedy rule did not foresee a viral clearance model that bears on a decision. The XMuLV model at low-pH viral inactivation is such a model. On it the step claim is not

met over the NOR as characterized, and the low-pH viral inactivation report (PCR-006) moves the NOR (Section 9.2). Under rule 2, which refits every viral clearance model and spikes no virus, the XMuLV model at low-pH viral inactivation was refitted, because viral clearance is not tested on the drug substance.

The refit removes terms by hierarchical backward elimination at $\alpha = 0.05$, and it is corrective action CA2. At each step, the least significant term that no remaining term contains is removed. A main effect therefore stays while an interaction or a square of it remains, even where the main effect itself is not significant. Elimination stops when every removable term is significant. In an augmented design the block term belongs to the design and is never removed. The refitted model is then judged against the three tests in the same way as the full model. Model selection, corrective action CA3, later replaced the refit by backward elimination for every output, the outputs that CA2 had refitted included (Section 8). The sample-size calculation rests on the refits.

Where neither the full nor the refitted model is adequate, the response is judged on its measured results, as the master plan asks for a response with no adequate model. Every measured result of the screening and response-surface designs of the step is compared with the criterion, at manufacturing scale where the scale-down qualification gives an offset for the output. The designs span the characterized range, so a response whose measured results all meet the criterion is taken to meet it over the NOR. A modelled output must instead meet its criterion with the 95 % prediction interval of a single batch at every combination of NOR low, set-point and NOR high. Section 8.5.2 gives the margin of each such response.

Table 5.1 gives the remedy of each failing model and the terms of each refit. A main effect in parentheses is not itself significant and is kept because an interaction or a square of it remains. The remedy rule gives 3 models an augmentation, 2 at the production bioreactor and 1 at anion exchange. Of the 11 refitted models, 5 are models of viral clearance, 3 have no effect to resolve and 3 have an effect but do not reach the target probability of success within the size of their planned design. Of the viral clearance models, the anion exchange models of XMuLV clearance and MVM clearance, and the virus filtration model of MVM clearance would reach the target with 18, 22 and 8 further runs. No runs were made for them, because viral clearance is not tested on the drug substance.

Table 5.1: Remedy of each model that does not meet the model-adequacy criteria.

Step	Attribute	Terms of the refit (planned runs)[a)]	Runs needed	Remedy	Reason
Production bioreactor	High mannose (%)	Culture pH, culture temperature, (culture duration), CO2 partial pressure (pCO2), culture pH × culture duration, culture pH × CO2 partial pressure (pCO2)	16	Augmentation[b)]	Attribute of the drug substance specification, with an effect
	Acidic variants (%)	Culture pH, culture temperature, (culture duration), CO2 partial pressure (pCO2), culture pH × culture temperature, culture pH × culture duration	12	Augmentation[b)]	Attribute of the drug substance specification, with an effect
Protein A	Leached Protein A (ppm)	None, the mean only	–	Refit	No effect to resolve
	Fragments, LMW (%)	Elution buffer pH	–	Refit	Probability of success not reached within the size of the planned design
Low-pH viral inactivation	XMuLV clearance (log10)	Inactivation pH, hold time, temperature	–	Refit	Viral clearance, not a drug substance test
	Fragments, LMW (%)	(Inactivation pH), hold time, (temperature), hold time × temperature, inactivation pH^2, $hold\ time^2$	–	Refit	Probability of success not reached within the size of the planned design
Cation exchange	Fragments, LMW (%)	(Elution buffer pH), elution stop collect, elution buffer pH × elution stop collect	–	Refit	Probability of success not reached within the size of the planned design
Anion exchange	Host cell protein (ng/mg)	Load pH, equil/Wash-1 conductivity, load pH × equil/Wash-1 conductivity, $equil/Wash\text{-}1\ conductivity^2$	12	Augmentation[b)]	Attribute of the drug substance specification, with an effect
	XMuLV clearance (log10)	Load pH, load conductivity, (protein load), load pH × protein load	18	Refit	Viral clearance, not a drug substance test
	MVM clearance (log10)	Load pH, load conductivity	22	Refit	Viral clearance, not a drug substance test
	Fragments, LMW (%)	None, the mean only	–	Refit	No effect to resolve
Virus filtration	MVM clearance (log10)	Filtration volume (load)	8	Refit	Viral clearance, not a drug substance test
	XMuLV clearance (log10)	None, the mean only	–	Refit	Viral clearance, not a drug substance test
	Fragments, LMW (%)	None, the mean only	–	Refit	No effect to resolve

[a)] Terms significant at $\alpha = 0.05$. A main effect in parentheses is not significant and is kept because an interaction or a square of it remains in the refit.

[b)] The step's augmentation has the largest number of runs that any of the models the rule augments at that step needs.

Runs needed: the smallest augmentation with a probability of success of at least 0.90, at the effects of the planned runs, that the planned model meets the lack-of-fit test and the limit on overfitting. A dash means not reached within the size of the planned design, or not computed when the refit keeps no effect. For a refitted model the number is what an augmentation would have needed, and no runs were made.

The refits of the fragments model of low-pH viral inactivation and the fragments model of cation exchange are not adequate either. Their refits keep effects that the design cannot predict well, with R^2 minus predicted R^2 of 0.27 and 0.21. Fragments at these two steps were therefore judged on their measured results, and model selection gives the same outcome for both (Section 8.5.2).

6 Sample-size calculation

6.1 Method

The number of further runs is calculated by simulation, separately for each failing model with an effect. The three tests of the model-adequacy criteria (PCMP-001 §7.2) that this investigation applies require a significant model F-test ($\alpha = 0.05$), no significant lack of fit ($\alpha = 0.05$) and R^2 minus predicted R^2 of at most 0.20. The simulation estimates the probability of success, which is the probability, at the effects estimated from the planned runs, that the planned model meets the second and the third of these tests once the design has been augmented by a given number of runs. It does not include the uncertainty of those estimates. The planned model is the full quadratic with a block term, on the raw scale, because that was the model the plans prescribed when the runs were sized. Model selection, and with it the log scale for some outputs, came later. The calculation has six steps:

1. The refit of the planned runs by backward elimination, with all its terms (Table 5.1), is taken as the true model, with its fitted coefficients. Its residual standard deviation is taken as the variation between runs. The refit is used because it keeps only the terms that the planned runs support. The full model would carry its fitted noise into every simulated data set.
2. For a candidate size, the D-optimal augmentation of that many runs is added to the planned runs as a second block (Section 6.2).
3. Responses are simulated at the planned and the augmentation runs, as the true model plus an error drawn from a normal distribution with the residual standard deviation, the same at every level. For each size, 4,000 data sets are simulated.
4. The planned model, the full quadratic with a block term, is fitted to each simulated data set. It is judged against the lack-of-fit test at $\alpha = 0.05$ and the limit of 0.20 on R^2 minus predicted R^2, the two tests that the remedy of the master plan names. The model F-test is not simulated. Pure error comes from the replicated centre points of the planned design, because no augmentation repeats a point within its own block.
5. The probability of success is estimated as the share of simulated data sets in which the planned model meets both tests.
6. Sizes are evaluated in steps of 2 runs, up to the size of the planned design. The number of runs needed is the smallest size with a probability of success of at least 0.90.

The simulation also gives the power of each significant effect of the refit. The power is the share of simulated data sets in which the term is significant at $\alpha = 0.05$ in the fitted planned model, the full quadratic with a block term. The probability of success does not always rise with every further run, because the D-optimal augmentation of

each size is a different set of points. In 9 of the 10 curves it falls at least once from one size to the next (Figure 6.1).

The probability of success cannot rise much above 0.95. The lack-of-fit test at α = 0.05 rejects a correct model in 5 % of data sets, whatever the size of the design. A target of 0.90 therefore asks that the limit on overfitting be met in about 95 % of data sets.

The calculation assumes that the refit of the planned runs is the true model, that the error is normal with the same variance at every level, and that the two blocks differ by a constant at most. A constant difference between the blocks is taken up by the block term of the fitted model. A change in the effects between the blocks is not. The augmentations repeat some corner points of the planned designs, and Section 8.5.1 compares the results there. On the planned runs that the simulation used, model selection put 2 of the outputs simulated here on the log scale (acidic variants at the production bioreactor and host cell protein at anion exchange), and on all runs also high mannose at the production bioreactor. For them, the error of constant variance on the raw scale is an assumption that the log-scale analysis does not support (Section 8.2). An effect that the planned runs did not detect is not in the true model, so the calculation does not size a design to find it. Each curve uses its own stream of a pseudo-random number generator with a fixed seed, so the calculation gives the same result each time it is run.

The calculation has two further limitations. First, it simulates the planned runs again together with the augmentation runs, instead of holding the planned results at their measured values. It therefore estimates how often a new design of planned and augmentation runs would meet the tests, not how often the measured planned results would meet them together with new runs. Second, the terms of the true model were kept because they were significant on the planned runs. Coefficients kept for that reason tend to lie further from zero than the effects they estimate, so the probability of success is optimistic.

6.2 D-optimal augmentation

Each augmentation is drawn from the levels of the face-centred design, so every factor takes its low, centre or high level. For four factors this gives 81 candidate points. The runs are chosen to maximise the determinant of the information matrix of the planned model, the full quadratic with a block term, with the planned runs held fixed in the first block. The selection adds, one run at a time, the candidate that increases the determinant most. It then exchanges chosen runs with candidates until no exchange increases the determinant further, which is a Fedorov exchange (Fedorov 1972). Ties go to the first candidate in a fixed order, so the design needs no random number.

The face-centred designs of three and four factors hold corner, axial and centre points only, and none of their points has exactly one factor at its centre level. Their squared terms are therefore strongly correlated with each other, because every corner run sets all of them to one. Both D-optimal augmentations consist mostly of points at the middle of an edge of the design cube, with one factor at its centre level and

the others at a limit (Section 7). These points reduce the correlation between the squared terms. In the production bioreactor design the largest correlation between two squared terms falls from 0.69 on the planned runs to 0.54 on all runs.

6.3 Results

Table 6.1 gives the probability of success of each failing model with an effect, with the planned runs alone and with further runs.

Table 6.1: Probability of success that the planned model meets the lack-of-fit test and the limit on overfitting, by the number of further runs.

Step	Attribute	Probability of success with further runs								Runs needed	Lowest power
		0	4	8	12	16	20	24	28		
Production bioreactor	High mannose (%)	0.251	0.344	0.543	0.722	0.909	0.934	0.946	0.954	16	0.76
	Acidic variants (%)	0.444	0.604	0.790	0.900	0.951	0.948	0.954	0.948	12	0.67
Protein A	Fragments, LMW (%)	0.001	0.000	0.000	0.000	0.000	0.000	0.000	0.001	–	–
Low-pH viral inactivation	XMuLV clearance (log10)	0.101	0.117	0.289	0.437	0.677	–	–	–	–	–
	Fragments, LMW (%)	0.105	0.101	0.209	0.455	0.448	–	–	–	–	–
Cation exchange	Fragments, LMW (%)	0.003	0.000	0.001	0.001	0.001	0.002	0.002	0.005	–	–
Anion exchange	Host cell protein (ng/mg)	0.493	0.629	0.800	0.921	0.951	0.949	0.950	0.946	12	0.55
	XMuLV clearance (log10)	0.222	0.240	0.403	0.696	0.874	0.915	0.949	0.947	18	0.91
	MVM clearance (log10)	0.194	0.184	0.311	0.624	0.814	0.895	0.945	0.946	22	1.00
Virus filtration	MVM clearance (log10)	0.433	0.808	0.929	0.952	–	–	–	–	8	1.00

Probability of success: the probability, at the effects estimated from the planned runs, that the planned model meets the lack-of-fit test and the limit on overfitting of PCMP-001 §7.2, estimated from 4,000 simulated data sets. It does not include the uncertainty of those estimates. The planned model is the full quadratic, with a block term where there are further runs. The data are simulated on the raw scale from the refit of the planned runs and its residual standard deviation. Further runs: a D-optimal augmentation of that size. Sizes in steps of 2 were computed, and the table shows every second size. A dash in a probability column marks a size larger than the planned design, which was not computed. Runs needed: the smallest size with a probability of success of at least 0.90, and a dash where it is not reached within the size of the planned design. Lowest power: the power, at that size and $\alpha = 0.05$, of the weakest significant effect of the refit. The lack-of-fit test rejects a correct model in 5 % of data sets, so no size takes the probability much above 0.95.

The three models that the remedy rule augments reach a probability of success of 0.90 within the size of their planned design. High mannose needs 16 further runs, with a probability of success of 0.909. Acidic variants need 12, with 0.900. The production bioreactor design is therefore augmented by 16 runs, the larger of the two, and at that size the probability of success for acidic variants is 0.951. Host cell protein at anion exchange needs 12 further runs, with 0.921, and the anion exchange design is augmented by that number.

Figure 6.1 shows the same curves at every size computed, with the target as a dashed line. The curves of the other failing models fall into three patterns. The curves of the anion exchange models of XMuLV clearance and MVM clearance, and the virus filtration model of MVM clearance cross the target within the size of the planned design. These models are refitted under rule 2 of the remedy rule, because viral clearance is not tested on the drug substance. The two models of low-pH viral inactivation rise with further runs but stay below the target up to the 18 runs of the planned design. At that size their probability of success is 0.754 for XMuLV clearance and 0.743 for fragments, which Figure 6.1 shows and Table 6.1, with every second size computed, does not. The fragment models of Protein A and cation exchange stay at or below 0.005 at every size. Their effects are too small against the residual variation for the full quadratic model to meet the lack-of-fit test and the limit on overfitting at any size computed.

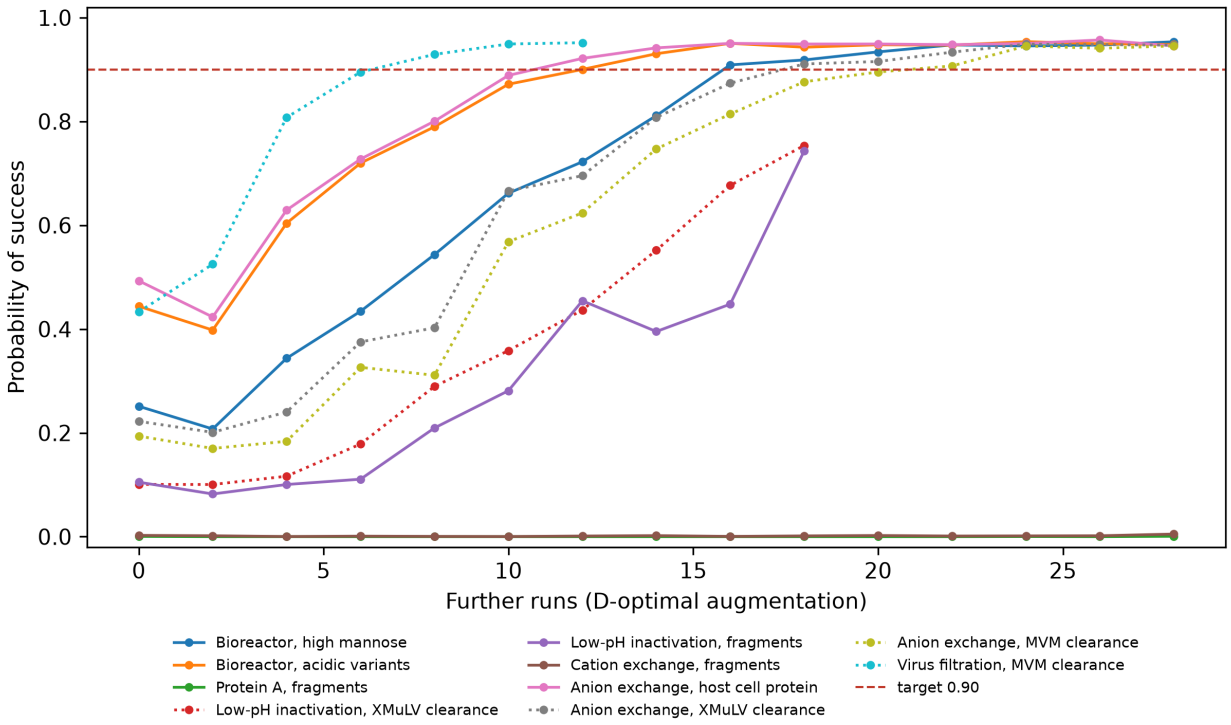


Figure 6.1: Probability of success that the planned model (full quadratic with a block term) meets the lack-of-fit test and the limit on overfitting of the model-adequacy criteria of PCMP-001, against the number of further runs in a D-optimal augmentation. One line per failing model with an effect. Viral clearance models are dotted. The dashed line is the target.

6.4 Power of the effects

The augmentations are sized for the lack-of-fit test and the limit on overfitting, not for the power of each effect. Table 6.2 gives the power of each significant effect of the refit at the size of the augmentation that was run. The residual standard deviations of the refits behind these figures are 0.43 percentage points for high mannose, 1.50 percentage points for acidic variants and 2.37 ng/mg for host cell protein. Of the 14 effects, 9 have a power of at least 0.90. The weakest is the term equil/Wash-1 conductivity² in the host cell protein model of anion exchange, with a power of 0.55. A non-significant interaction or curvature term in a re-evaluated model is therefore consistent with a small effect. It is not evidence that the effect is absent.

Table 6.2: Power of each significant effect of the refit at the size of the step's augmentation.

Step	Attribute	Effect	Power
Production bioreactor	High mannose (%)	Culture pH	1.00
		Culture temperature	1.00
		CO2 partial pressure (pCO2)	0.76
		Culture pH × culture duration	0.89
		Culture pH × CO2 partial pressure (pCO2)	0.93
	Acidic variants (%)	Culture pH	1.00
		Culture temperature	1.00
		CO2 partial pressure (pCO2)	1.00
		Culture pH × culture temperature	0.96
		Culture pH × culture duration	0.69
Anion exchange	Host cell protein (ng/mg)	Load pH	1.00
		Equil/Wash-1 conductivity	1.00
		Load pH × equil/Wash-1 conductivity	0.77
		Equil/Wash-1 conductivity²	0.55

Power is the share of 4,000 simulated data sets in which the term is significant at α = 0.05 in the planned model, the full quadratic with a block term, with 16 further runs at the production bioreactor and 12 at anion exchange. Only the significant terms of each refit are listed. The lowest power column of the table of the probability of success by the number of further runs gives the power at the size each model needs on its own.

7 Augmentation designs

Each augmentation was executed as a second block of its response-surface design, with the planned runs as the first block. The augmentation runs continue the run numbers of the planned design. They were executed in a randomized order, as the process characterization master plan requires (PCMP-001 §6.1), and the design tables give them in the order of the design. The two designs were approved on 2026-09-02 with the remedy rule.

7.1 Production bioreactor

The production bioreactor augmentation has 16 runs (Table 7.1). They were made in study S-BRX-AUG, from 2026-09-03 to 2026-09-24. Of these runs, 11 lie at the middle of an edge of the design cube and 5 at a corner. The corner runs repeat settings that the planned factorial runs also used. Every run was tested for all 5 responses of the design (afucosylation, galactosylation, high mannose, acidic variants and aggregates). The production bioreactor report (PCR-003) therefore re-evaluates every model of the step on all runs, including the 3 models that met the three tests of the model-adequacy criteria (a significant model F-test ($\alpha = 0.05$), no significant lack of fit ($\alpha = 0.05$) and R^2 minus predicted R^2 of at most 0.20) on the planned runs.

Table 7.1: D-optimal augmentation of the production bioreactor response-surface design.

Run	Coded level				Setting			
	A	B	C	D	A, Culture pH	B, Culture temperature (°C)	C, Culture duration (days)	D, CO2 partial pressure (pCO2, mmHg)
29	-1	-1	-1	+1	6.60	34	15	160
30	-1	-1	0	-1	6.60	34	17	40
31	-1	-1	+1	0	6.60	34	19	100
32	-1	0	+1	+1	6.60	35	19	160
33	-1	+1	-1	-1	6.60	36	15	40
34	-1	+1	0	+1	6.60	36	17	160
35	-1	+1	+1	-1	6.60	36	19	40
36	0	-1	-1	-1	6.85	34	15	40
37	0	-1	+1	+1	6.85	34	19	160
38	0	+1	-1	+1	6.85	36	15	160
39	+1	-1	-1	0	7.10	34	15	100
40	+1	-1	0	+1	7.10	34	17	160
41	+1	-1	+1	-1	7.10	34	19	40
42	+1	0	-1	+1	7.10	35	15	160
43	+1	+1	0	-1	7.10	36	17	40
44	+1	+1	+1	+1	7.10	36	19	160

Factors: A, Culture pH. B, Culture temperature. C, Culture duration. D, CO2 partial pressure (pCO2).
Block 2. Runs are numbered on from the planned design.

7.2 Anion exchange

The anion exchange augmentation has 12 runs (Table 7.2). They were made in study S-AEX-AUG, from 2026-09-10 to 2026-09-24. Of these runs, 10 lie at the middle of an edge of the design cube and 2 at a corner, and the corner runs repeat planned factorial settings. Every run was tested for host cell protein, step yield and fragments by non-reduced CE-SDS. No virus was spiked, so XMuLV and MVM clearance were not measured, and their models keep the planned runs alone.

Table 7.2: D-optimal augmentation of the anion exchange response-surface design.

Run	Coded level				Setting			
	A	B	C	D	A, Load pH	B, Equil/Wash-1 conductivity (mS/cm)	C, Load conductivity (mS/cm)	D, Protein load (g/L resin)
29	-1	-1	-1	0	7.2	1.6	3.0	175
30	-1	-1	0	-1	7.2	1.6	5.5	50
31	-1	0	+1	+1	7.2	2.6	8.0	300
32	-1	+1	0	+1	7.2	3.6	5.5	300
33	-1	+1	+1	-1	7.2	3.6	8.0	50
34	0	-1	-1	-1	7.5	1.6	3.0	50
35	0	+1	-1	+1	7.5	3.6	3.0	300
36	+1	-1	+1	+1	7.8	1.6	8.0	300
37	+1	0	-1	+1	7.8	2.6	3.0	300
38	+1	0	+1	-1	7.8	2.6	8.0	50
39	+1	+1	-1	0	7.8	3.6	3.0	175
40	+1	+1	0	-1	7.8	3.6	5.5	50

Factors: A, Load pH. B, Equil/Wash-1 conductivity. C, Load conductivity. D, Protein load.

Block 2. Runs are numbered on from the planned design.

8 Model selection (corrective action CA3)

8.1 Procedure

Model selection was adopted as corrective action CA3 on 2026-09-15, before the investigation into acceptance criteria set after the data were seen (INV-001) closed on 2026-09-18. It replaced the analysis of every output, the refits of corrective action CA2 included. Each report with a response-surface design applies it to every output of the design, on all runs in which the output was measured. Where a design was augmented, the planned runs form the first block and the augmentation runs the second. The verification of the platform worst cases behind the intermediate acceptance criteria applies the same procedure (Section 8.4). The process characterization master plan (PCMP-001) is not changed. Each report records the model selection as a discrepancy from its plan (Section 9). The procedure has four steps:

1. The scale. The output is raised to a power λ, and the full quadratic model is fitted to the result, with the block term where the runs form two blocks, over a range of powers (Box and Cox 1964). A power of λ = 1 leaves the output on its raw scale, and λ = 0 is read as the natural logarithm. For each power the profile likelihood measures how well the model fits the transformed output with residuals that are normal and of equal spread. Its 95 % interval is the set of powers that the data do not reject at the 5 % level. The output is modelled on the log scale (λ = 0) where this interval contains 0 and excludes 1. Otherwise it is modelled on the raw scale (λ = 1). The rule chooses between these two scales only, and only an output whose results are all positive can take the log scale. Where the interval excludes both 0 and 1, the raw scale is kept. On all runs this applies to afucosylation at the production bioreactor. On its planned runs alone the interval of afucosylation contains 0 and excludes 1, and the model selected there is on the log scale. The scale is chosen on the full quadratic before the search, so that the search compares every candidate on one scale. An output whose results the full quadratic reproduces exactly has no interval, because its profile likelihood is then unbounded, and no scale is chosen for it.
2. The search. Every submodel of the full quadratic model that respects strong hierarchy is fitted on that scale. Strong hierarchy allows an interaction only together with both of its main effects, and a squared term only together with its main effect. The block term is kept in every candidate. The candidate with the lowest AICc is selected (Section 8.3). Where the full quadratic reproduces every run exactly, AICc is not defined, and the output has no adequate model (step 3).
3. The check. The selected model is judged on its own scale against the three tests of the model-adequacy criteria of the master plan (Section 3), which require a significant model F-test (α = 0.05), no significant lack of fit (α = 0.05) and R^2

minus predicted R² of at most 0.20. The fourth criterion, on the pattern of the residuals and on influential runs, is judged by the report of the step. A selected model with no term has no model F-test and is judged by the other two tests. A model that reproduces every run exactly is not adequate. With no residual there is no pure error for the lack-of-fit test and no residual mean square for the model F-test, so the criteria cannot be applied, and its prediction interval would have zero width. Only the selected model is judged. If it fails, no other candidate takes its place. The selection used the same runs that the three tests then judge. A model chosen as the best of many candidates fits those runs better than a model fixed in advance, so the check of a selected model against the three tests is optimistic. Section 8.5.1 gives the one independent check that exists.

4. The offset. The scale-down offset of an output is taken on the scale of its model, with its 90 % confidence interval. For a raw-scale output it is the mean of the scale-down set-point runs minus the mean of the manufacturing-scale batches, and it is subtracted from a scale-down prediction. For a log-scale output it is the same difference of the mean logarithms, reported as a ratio of geometric means, and a scale-down prediction is divided by it. An output with no adequate model takes its offset on the scale chosen in step 1, or on the raw scale where no scale is chosen.

The check of a selected model against the three tests has three outcomes. A selected model with terms that passes is the predictive model of the output, and it sets the ranges of its step. A selected model with no term passes when it meets the lack-of-fit test and the limit on overfitting. It says that no effect of the parameters was detected within the range studied. It does not show that there is none. For a model of the mean alone on a single block, R² is zero and R² minus predicted R² depends on the number of runs only, 0.19 for 12 runs and 0.08 for 28, so for such a model the check reduces to the lack-of-fit test. With 3 degrees of freedom for pure error, that test has little power, and such a model passes it almost always. An output whose selected model fails, or fits exactly, has no adequate model. It is judged on its measured results, as Section 5 describes, and it sets no range.

8.2 Why a log scale, and what it changes

Host cell protein in the Protein A pool shows what the log scale does. Every model of pool host cell protein in this section puts response-surface run 23 at the elution buffer pH recorded under study event DEV-005-01, as the Version 2.0 analysis does (discrepancy PD-005-03). On the log scale the selected model has the form

$$\ln y = b_0 + b_1 x_1 + b_2 x_2 + \cdots + \varepsilon$$

and, transformed back,

$$y = e^{b_0} \cdot e^{b_1 x_1} \cdot e^{b_2 x_2} \cdots e^{\varepsilon}$$

Here y is pool host cell protein in ng/mg, x_1, x_2 and so on are the parameters in coded units, from −1 at the low end of the range studied to +1 at the high end, and ε is the error of a single run. One coded unit is therefore half the range studied. A parameter that enters the model as a main effect alone multiplies the output by a fixed factor, e^b per coded unit and e^{2b} from the low to the high end of its range. The error multiplies it too, so the scatter in ng/mg grows with the level.

Protein load shows how this reads in real units. Its range studied is 10 to 50 g/L resin, so one coded unit is 20 g/L resin. Its coefficient on the log scale is b = 0.443. Pool host cell protein therefore rises by a factor of $e^b = 1.56$ for every 20 g/L resin of protein load, and by $e^{2b} = 2.43$ from 10 to 50 g/L resin. Across the range studied, the selected model multiplies pool host cell protein by a factor of 2.43 for protein load, 0.30 for elution buffer pH and 0.85 for end of pool collect, from the low to the high end of each parameter. A factor below one is a decrease. Load flow rate also has a squared term, so its factor depends on where along the range the change is made. From 100 to 200 cm/hr pool host cell protein changes by a factor of 0.88, and from 200 to 300 cm/hr by a factor of 1.08.

Two properties of the process and the method make host cell protein behave this way. A purification step removes a share of what is present, so a parameter changes the share that is removed, not a fixed amount. The precision of the host cell protein ELISA (AMV-3012) is a percentage of the level, 9.5 %, so the scatter of the results grows with the level. Effects that multiply look like an interaction when they are added on the raw scale. On the raw scale the full quadratic model needs the interaction protein load × elution buffer pH ($p < 0.001$). On the log scale the same interaction is not significant ($p = 0.156$), and the selected model has no interaction at all.

Least squares needs no assumption about the distribution of the residuals to estimate the coefficients. The t-tests, F-tests and prediction intervals built on it assume residuals that are independent and normally distributed, with mean zero and the same variance over the whole range studied. The residuals of the full quadratic model of pool host cell protein show how the raw scale departs from this. Its runs were split at the median of the predicted values, 14 runs in the lower half and 14 in the upper half. On the raw scale the standard deviation of the residuals is 638 ng/mg in the lower half of the predictions and 947 ng/mg in the upper half, 1.48 times as large. On the log scale the same split gives 0.067 and 0.074 in natural-log units, 1.11 times as large. On the log scale, therefore, the spread no longer grows with the level. Two halves of this size cannot show on their own that the raw-scale spread differs, and the Box-Cox profile is the evidence for the scale. It puts λ at 0.06, with a 95 % interval from −0.32 to 0.42. The interval contains 0 and excludes 1, so the data support the log scale and reject the raw scale. This evidence concerns the full quadratic model, on which the scale is chosen, and not the smaller model selected afterwards. A test can fail to reject the assumption of equal variance, but it cannot show that the assumption holds.

The log scale changes the following:

1. The criterion stays in its own unit, ng/mg, and the prediction limits are transformed back into it.
2. Effects are fold changes, as above.

3. The scale-down offset is a ratio of geometric means, 0.97 for pool host cell protein with a 90 % confidence interval from 0.89 to 1.06, and a scale-down prediction is divided by it.
4. A prediction interval is calculated on the log scale and transformed back. It keeps its 95 % coverage, because the probability that ln y lies between two values is the probability that y lies between their antilogs.
5. The interval is asymmetric, further above the predicted value than below it. At the least favourable combination of the NOR in force (protein load 39 g/L resin, elution buffer pH 3.4, load flow rate 150 cm/hr and end of pool collect 2.3 CV), the model predicts ln y = 9.382 in the scale-down model. The 95 % prediction interval of a single batch is two-sided. On the log scale it is the prediction ± $t \cdot s \cdot \sqrt{(1 + h)}$, where t = 2.074 is the 97.5th percentile of the t distribution with 22 degrees of freedom, s = 0.105 is the residual standard deviation on the log scale, and h = 0.107 is the leverage of this combination. This gives 9.382 ± 0.230. Transformed back, the interval runs from 9,440 to 14,941 ng/mg around a median of 11,876 ng/mg in the scale-down model. Divided by the scale-down offset of 0.97, it runs from 9,698 to 15,349 ng/mg around a median of 12,201 ng/mg at manufacturing scale, 2,503 below and 3,148 above. The conclusion over the NOR uses this two-sided interval. The verification of the platform worst cases uses the one-sided 95 % bound instead (Section 8.4).
6. The predicted value is the median of the output, its geometric mean, not its arithmetic mean. The upper prediction limit, which is judged against an upper criterion, is not affected by that.
7. Near an upper criterion, where the level is high, a raw-scale model pools one residual standard deviation over all levels and so understates the scatter, which the log-scale interval does not.

In the reports, 7 of the 22 outputs with a criterion are modelled on the log scale. They are high mannose, acidic variants and aggregates at the production bioreactor, host cell protein in the Protein A pool, aggregates and host cell protein at cation exchange, and host cell protein at anion exchange.

8.3 How the model is searched

The goal of the search is the model with the lowest AICc among all hierarchical sub-models of the full quadratic. AICc estimates how well a model would predict new runs (Hurvich and Tsai 1989; Burnham and Anderson 2002). AICc is the sum of a fit term, a penalty and a correction for small samples. The fit term is minus twice the maximized log-likelihood of the model with normally distributed errors. It equals $n \cdot \ln(SSE/n) + n(1 + \ln 2\pi)$, where n is the number of runs and SSE the residual sum of squares, and it falls as the fit improves. The penalty is 2p, where p is the number of coefficients, the intercept included. The correction is $2k(k + 1)/(n - k - 1)$, where $k = p + 1$ counts the coefficients and the residual variance. A term is therefore kept only where it lowers the fit term by more than it raises the penalty and the correction. AICc is Akaike's criterion, AIC, with that correction. The correction raises the penalty when the runs are few for the number of terms, as in these designs, so that

the criterion does not favour models with too many terms. The part n(1 + ln 2π) of the fit term is the same for every candidate of an output, so only the differences in AICc between candidates carry information. For the full quadratic model on 28 runs, p is 15, so the penalty is 30, and k is 16, so the correction adds 49.5.

Pool host cell protein in the Protein A pool shows the trade, with run 23 at its recorded pH as in Section 8.2. On the log scale the fit term is −69.2 for the full quadratic model and −53.4 for the selected model. The selected model has 5 terms besides the intercept, so p is 6 and its penalty is 12, and k is 7, so its correction is 5.6. The full quadratic fits better by 15.8, but its penalty and correction together are larger by 61.9. The selected model therefore has the lower AICc, −35.8 against 10.3 for the full quadratic model, a difference of 46.1.

The search fits every candidate, 1,337 for a design with four factors, 95 for three and 13 for two, the model of the mean included. It takes the candidate with the lowest AICc. The candidates are listed in a fixed order, from the fewest main effects upward, and a tie goes to the first. A stepwise search, by contrast, adds or removes one term at a time and stops when no single change lowers AICc. It therefore stops where only a combination of changes would lower AICc further.

On this campaign’s data the two searches were compared for 21 outputs with a criterion. The output that fits exactly has no AICc and is left out. A backward stepwise search from the full quadratic would have stopped at a model with a higher AICc than the best candidate for 9 of them, a forward search from the mean for 6, and one or the other for 12. For host cell protein at cation exchange, for example, the exhaustive search selects 6 terms with an AICc of −31.2. A backward search stops at 8 terms (−27.4) and a forward search at 3 terms (−27.9).

The backward elimination of corrective action CA2 differed from this search in three ways. It was stepwise. It decided by the p-value of each term. That p-value compares the model with and without the term, but it does not weigh the fit against the number of terms, and it cannot compare models that are not nested. It also worked on the raw scale only.

8.4 Verification of the platform worst cases

The process characterization master plan (PCMP-001) Version 2.0 derives the intermediate acceptance criteria a priori. Each drug substance criterion is carried back through the platform worst case of every downstream step, its lowest clearance fold or its largest increase, and each platform worst case is the result of a single run. The verification of these worst cases uses the same model selection, on the planned runs of each downstream step. For each worst case it predicts the output of the step at every combination of NOR low, set-point and NOR high of the NOR as characterized. At each combination it also takes the one-sided 95 % prediction bound of a single batch, on the scale of the model and transformed back. Both are converted into a clearance fold or an increase, and the least favourable value over the combinations is the bound of the step.

Where the predicted value of a step is worse than the platform worst case, the bound replaces the platform value, and the criteria upstream of the step tighten. Where the bound itself is better than the platform worst case, on a selected model that meets the three tests of the model-adequacy criteria (a significant model F-test ($\alpha = 0.05$), no significant lack of fit ($\alpha = 0.05$) and R^2 minus predicted R^2 of at most 0.20) and is not an exact fit, the bound replaces the platform value, and the criteria upstream widen. Otherwise the platform value stays. Both directions use the one-sided 95 % prediction bound of a single batch, because every output is judged per batch and the platform worst case that the bound replaces was a single run. A bound from a tolerance interval covers a stated share of all batches with a stated confidence, and it is wider. The bound is pointwise at each combination, not simultaneous over them.

Under the master plan the fourth criterion, on the residuals and on influential runs, does not decide a widening, and a bound is examined instead for whether it depends on a single run (PCMP-001 §7.5). Neither of the two bounds behind the widenings depends on a single run. For each bound, every run whose Cook's distance exceeds 4/n, with n the number of runs, was deleted in turn, and the model selection and the bound were computed again. These are runs 2 and 9 for host cell protein at anion exchange and runs 3, 7 and 8 for aggregate at low-pH viral inactivation. Each time the selected model met the three tests and the bound stayed better than the platform value, from 4.33 to 4.56 against a fold of 4, and from 0.546 to 0.551 against an increase of 0.70 percentage points.

The platform worst cases of the low-pH hold are increases over a hold of 180 min. The A-Mab value is the increase within the NOR as characterized, which allows a hold of at most 120 min. At low pH the antibody forms aggregate, and the aggregate grows with the time at that pH. At the edges of the range studied, which include a hold of 180 min, A-Mab adds up to 0.84 percentage points of aggregate, more than the platform value of 0.70. Its single-batch bound of 0.559 within the NOR as characterized is smaller than the platform value because that NOR ends at a shorter hold. The selected model of aggregate at low-pH viral inactivation has terms in hold time and temperature only, and none in inactivation pH. The NOR in force ends at the same longest hold time, 120 min, and the same highest temperature, 23 °C, as the NOR as characterized. The bound over the NOR as characterized therefore also covers the NOR in force. For acidic variants the model at low pH fits exactly and is not adequate, so its bound replaces no platform value (Section 8.5.3). Section 9.1 gives the criteria that the bounds of the selected models changed, and the risk of each change.

8.5 What model selection changed

Model selection gives an adequate model for 18 of the 22 outputs with a criterion. Every output whose full quadratic model met all three tests has an adequate selected model. Of the 14 outputs whose full quadratic model failed, 11 have an adequate selected model and 3 have none. The output whose full quadratic model fits exactly, acidic variants at low-pH viral inactivation, has none either. Table 8.1 gives the verdict for every selected model, and Table 9.1 gives the model each report uses.

Table 8.1: Selected model of every response-surface output with a criterion against three of the model-adequacy criteria of PCMP-001.

Step	Attribute	Scale	Terms	Runs	F-test p	Lack-of-fit p	R^2	Predicted R^2	R^2 minus predicted R^2	Verdict
Production bioreactor	Afucosylation (%)	Raw	9	44	<0.001	0.246	0.95	0.92	0.04	adequate
	Galactosylation (%)	Raw	7	44	<0.001	0.994	0.96	0.94	0.02	adequate
	High mannose (%)	Log	3	44	<0.001	0.820	0.82	0.77	0.05	adequate
	Acidic variants (%)	Log	9	44	<0.001	0.295	0.94	0.90	0.05	adequate
	Aggregates, HMW (%)	Log	4	44	<0.001	0.514	0.95	0.94	0.02	adequate
Protein A	Host cell protein (ng/mg)	Log	5	28	<0.001	0.088	0.98	0.96	0.01	adequate
	Leached Protein A (ppm)	Raw	0	28	–	0.907	0.00	−0.08	0.08	adequate
	Fragments, LMW (%)	Raw	2	28	0.002	0.776	0.40	0.24	0.16	adequate
Low-pH viral inactivation	Aggregates, HMW (%)	Raw	2	18	<0.001	0.504	0.99	0.99	0.00	adequate
	XMuLV clearance (log10)	Raw	3	18	<0.001	0.546	0.79	0.67	0.12	adequate
	Acidic variants (%)	–[a)]	1	18	n.t.	n.t.	1.00	1.00	0.00	not testable: exact fit
	Fragments, LMW (%)	Raw	4	18	0.003	0.403	0.68	0.35	0.33	not adequate
Cation exchange	Aggregates, HMW (%)	Log	8	28	<0.001	0.467	0.97	0.94	0.03	adequate
	Host cell protein (ng/mg)	Log	6	28	<0.001	0.846	0.95	0.92	0.03	adequate
	Fragments, LMW (%)	Raw	3	28	0.001	0.712	0.48	0.27	0.21	not adequate
Anion exchange	Host cell protein (ng/mg)	Log	2	40	<0.001	0.842	0.90	0.88	0.02	adequate
	XMuLV clearance (log10)	Raw	4	28	<0.001	0.928	0.86	0.80	0.06	adequate
	MVM clearance (log10)	Raw	2	28	<0.001	0.747	0.84	0.80	0.04	adequate
	Fragments, LMW (%)	Raw	0	40	–	0.923	0.07	−0.04	0.11	adequate
Virus filtration	MVM clearance (log10)	Raw	1	12	<0.001	0.105	0.90	0.84	0.06	adequate
	XMuLV clearance (log10)	Raw	1	12	0.053	0.789	0.32	0.08	0.24	not adequate
	Fragments, LMW (%)	Raw	0	12	–	0.911	0.00	−0.19	0.19	adequate

[a)] No scale is chosen. The results have no scatter, so the full quadratic reproduces them at every power of the Box-Cox profile, whose likelihood is then unbounded.

Scale: the scale chosen by the Box-Cox profile of the full quadratic. Terms: the terms of the selected model besides the intercept and the block term. Runs: every run in which the output was measured, planned and augmentation runs. F-test p: the p-value of the model F-test, with a dash where the model has no term and so no F-test. n.t.: not testable, because the model reproduces every run and has no residual. Verdict: adequate when the model meets the three tests of PCMP-001 §7.2 that this investigation applies, a significant model F-test ($\alpha = 0.05$), no significant lack of fit ($\alpha = 0.05$) and R^2 minus predicted R^2 of at most 0.20, on the scale of the model. R^2 minus predicted R^2 is calculated before rounding. Protein A response-surface run 23 is at the elution buffer pH recorded under study event DEV-005-01.

8.5.1 The augmentations

The three models that the augmentations were made for meet the three tests of the model-adequacy criteria (a significant model F-test ($\alpha = 0.05$), no significant lack of fit ($\alpha = 0.05$) and R^2 minus predicted R^2 of at most 0.20) on the planned runs alone once they are selected. They are the models of high mannose and acidic variants at the production bioreactor and of host cell protein at anion exchange. The augmentations were therefore not needed to pass these tests. They were decided on 2026-09-02 under the model of the plan, the full quadratic, which failed the limit on overfitting on the planned runs. Both started before model selection was adopted on 2026-09-15. The anion exchange runs were still under way on that date, as were the production bioreactor runs, and both were completed.

The augmentations were sized so that the full quadratic model with a block term would meet the lack-of-fit test and the limit on overfitting on all runs. On all runs it meets both for acidic variants and for host cell protein at anion exchange, with R^2 minus predicted R^2 of 0.12 and 0.11. For high mannose it does not, because R^2 minus predicted R^2 is 0.25 against the limit of 0.20. Its lack-of-fit test is passed ($p = 0.638$).

Table 8.2 gives the three selected models on the planned runs and on all runs, with the statistics of the three tests. Each meets the three tests on both sets of runs. For high mannose, 4 terms of the model selected on the planned runs are not in the model selected on all runs. These are culture duration, CO2 partial pressure (pCO2), culture pH × culture duration and culture pH × CO2 partial pressure (pCO2). On all runs the selected model of high mannose is on the log scale and has 3 terms (culture pH, culture temperature and culture pH × culture temperature). The scale changed with the second block as well, so part of the change in terms may come from the scale. For acidic variants every term of the model selected on the planned runs is kept on all runs, and the second block resolves 4 further terms (culture duration, culture pH × culture duration, culture temperature × CO2 partial pressure (pCO2) and culture pH^2). For host cell protein at anion exchange the selected model is the same on both sets of runs. It has the main effects of load pH and equil/Wash-1 conductivity, on the log scale.

Table 8.2: Selected models of the outputs that the augmentations were made for, on the planned runs and on all runs.

Step	Attribute	Runs	Scale	Terms	R^2	Predicted R^2	R^2 minus predicted R^2	Lack of fit p
Production bioreactor	High mannose (%)	28	Raw	Culture pH, culture temperature, culture duration, CO2 partial pressure (pCO2), culture pH × culture duration, culture pH × CO2 partial pressure (pCO2)	0.88	0.80	0.08	0.839
		44	Log	Culture pH, culture temperature, culture pH × culture temperature	0.82	0.77	0.05	0.820
	Acidic variants (%)	28	Log	Culture pH, culture temperature, CO2 partial pressure (pCO2), culture pH × culture temperature, culture pH × CO2 partial pressure (pCO2)	0.91	0.83	0.08	0.180
		44	Log	Culture pH, culture temperature, culture duration, CO2 partial pressure (pCO2), culture pH × culture temperature, culture pH × culture duration, culture pH × CO2 partial pressure (pCO2), culture temperature × CO2 partial pressure (pCO2), culture pH^2	0.94	0.90	0.05	0.295
Anion exchange	Host cell protein (ng/mg)	28	Log	Load pH, equil/Wash-1 conductivity	0.90	0.88	0.02	0.824
		40	Log	Load pH, equil/Wash-1 conductivity	0.90	0.88	0.02	0.842

Runs: the 28 planned runs, or all runs with the augmentation as a second block. Scale: the scale chosen by the rule of step 1 of the procedure. Terms: the terms of the model with the lowest AICc, besides the intercept and the block term. The statistics of the three tests are calculated on the scale of the model. The model F-test of every model in the table is significant ($p < 0.001$).

Table 8.3 tests each model selected on the planned runs of the two augmented steps against the augmentation runs, which it has not seen. If a model is right, each new run falls inside its 95 % prediction interval with a probability of 0.95, and its prediction error is close to its own residual standard deviation. The effectiveness check of Section 11 judges such a count by a one-sided binomial test at the 0.05 level. With 16 new runs, that test rejects a model with fewer than 14 inside. Only 13 of the 16 new runs fall inside the interval for high mannose and 11 for aggregates at the production bioreactor, and the test rejects both models ($p = 0.043$ and $p < 0.001$). Their prediction error is 1.7 and 1.8 times the residual standard deviation of the model. For the other outputs the ratio is at most 1.3, and every count passes the test.

Table 8.3: Prediction of the augmentation runs by the models selected on the planned runs.

Step	Attribute	Terms, planned → all runs	Scale, planned → all runs	New runs inside 95 % interval	Prediction error / SD	Block p, all runs
Production bioreactor	Afucosylation (%)	9 → 9	Log → raw	14 of 16	1.30	0.201
	Galactosylation (%)	7 → 7	Raw → raw	15 of 16	1.11	0.497
	High mannose (%)	6 → 3	Raw → log	13 of 16	1.70	0.329
	Acidic variants (%)	5 → 9	Log → log	16 of 16	0.92	0.294
	Aggregates, HMW (%)	5 → 4	Log → log	11 of 16	1.81	0.382
Anion exchange	Host cell protein (ng/mg)	2 → 2	Log → log	12 of 12	0.91	0.526
	Fragments, LMW (%)	0 → 0	Raw → raw	11 of 12	1.30	0.091

New runs: the augmentation runs, which the model selected on the planned runs has not seen. Inside: the new runs inside its 95 % prediction interval for a single run. Prediction error / SD: the root mean square error of its predictions of the new runs, divided by its own residual standard deviation, both on the scale of the model. Block p, all runs: the p-value of the block term in the model selected on all runs, which is a different model from the one whose predictions the row tests.

The block term of the models on all runs tests for a constant shift between the planned and the augmentation runs. Its p-values give no evidence of one, the smallest being 0.091. That is not evidence that nothing changed between the two campaigns. The planned production bioreactor runs ended on 2025-12-12 and the augmentation began on 2026-09-03. A change in an effect would show at the corners that the augmentation repeats, runs 29, 33, 35, 41 and 44 at the production bioreactor, as unequal differences between the repeated and the planned results. For high mannose these differences vary no more than run-to-run variation explains ($p = 0.565$). For aggregates they vary more, though not significantly at the 5 % level ($p = 0.054$). With only 5 pairs the comparison has little power, so a change in the aggregate effects between the campaigns cannot be excluded.

On all runs, high mannose loses the interactions culture pH × culture duration and culture pH × CO_2 partial pressure (pCO_2) that the planned runs suggested. Aggregates lose the terms culture temperature and culture temperature^2 and gain a curvature in culture pH. The reports use the models of all runs. For acidic variants the model selected on the planned runs predicts every new run within its interval, and the model of all runs adds the terms that the second block resolves.

8.5.2 Outputs with no adequate model

After selection, 4 outputs have no adequate model. They are acidic variants at low-pH viral inactivation, fragments at low-pH viral inactivation, fragments at cation exchange and XMuLV clearance at virus filtration. Each is judged on its measured results (Section 5). Table 8.4 gives each criterion, the number of results, the most adverse measured result and the margin between them. Every measured result meets its criterion. None of these outputs sets a range.

Table 8.4: Outputs with no adequate model, judged on their measured results.

Step	Attribute	Criterion	Number of results	Most adverse result	Margin
Low-pH viral inactivation	Acidic variants (%)	18.0–40.0	29	20.2	2.21
	Fragments, LMW (%)	≤ 2.60	18	1.13	1.47
Cation exchange	Fragments, LMW (%)	≤ 2.00	28	0.76	1.24
Virus filtration	XMuLV clearance (log10)	≥ 4.62	19	5.90	1.28

Number of results: every measured result of the screening and response-surface designs of the step. Fragments have nrCE-SDS results in the response-surface designs only. Most adverse: the highest result against an upper criterion, the lowest against a lower one, and the result nearest either limit against a two-sided criterion. Margin: the distance from the most adverse result to the criterion, in the unit of the attribute. Acidic variants are taken to manufacturing scale by subtracting the scale-down offset. The scale-down qualification did not compare fragments or viral clearance at both scales, so these results have no offset and are given as measured.

The fragment models at low-pH viral inactivation and at cation exchange had no adequate refit under corrective action CA2 either. XMuLV clearance at virus filtration is the one output for which model selection and the refit by backward elimination disagree on whether an adequate model exists. The refit had reduced it to its mean, which meets the lack-of-fit test and the limit on overfitting. Model selection keeps a term in filtration pressure, because that model has the lower AICc (−7.8 against −6.8 for the mean alone). Only the selected model is judged. Its model F-test is not significant ($p = 0.053$), and its R^2 minus predicted R^2 of 0.24 exceeds the limit of 0.20. XMuLV clearance at virus filtration is therefore judged on its measured results. The two models differ by 1.0 in AICc. Models within two units of each other have about equal support (Burnham and Anderson 2002), so the outcome for this output rests on a small difference.

8.5.3 An output that fits exactly

Acidic variants at low-pH viral inactivation have results with no scatter. Every run at the same hold time returned the same value, in the screening and the response-surface designs alike, the replicated centre points included. The full quadratic model, which contains hold time and its square, therefore reproduces every run exactly, and so does the model with hold time alone. With no residual there is no pure error for the lack-of-fit test and no residual mean square for the model F-test, so the model-adequacy criteria cannot be applied. A prediction interval would have zero width. The Box-Cox profile likelihood is unbounded, so no scale is chosen and no interval is reported. The output therefore has no adequate model, and it is judged on its measured results (Table 8.4). Its bound cannot replace a platform value in the verification of the platform worst cases (Section 8.4). The upper limit of acidic variants at the production bioreactor therefore stays at its a priori value of 39.0 %.

8.5.4 Outputs with no detected effect

The selected models of 3 outputs have no term. They are leached Protein A in the Protein A pool, fragments at anion exchange and fragments at virus filtration. No effect of the parameters on them was detected within the ranges studied. The model is their mean, with the block term where the design has two blocks.

9 Impact assessment

Model selection, corrective action CA3, replaces the full quadratic model of every output with a criterion in the report of its step. Table 9.1 lists the outputs by report. For each output it gives the remedy that the remedy rule gave a full quadratic model that failed one of the three tests of the model-adequacy criteria (a significant model F-test ($\alpha = 0.05$), no significant lack of fit ($\alpha = 0.05$) and R^2 minus predicted R^2 of at most 0.20), the model that the report uses after model selection, and the discrepancies from the plan that the report records.

Table 9.1: Response-surface outputs with a criterion by report, with the remedy and the model the report uses.

Report	Attribute	Remedy	Model in the report	Discrepancies from the plan
PCR-003 Production bioreactor	Afucosylation	–	Raw, 9 terms, 44 runs	PD-003-02, augmentation PD-003-03, model selection
	Galactosylation	–	Raw, 7 terms, 44 runs	
	High mannose	Augmented (CA1)	Log, 3 terms, 44 runs	
	Acidic variants	Augmented (CA1)	Log, 9 terms, 44 runs	
	Aggregates	–	Log, 4 terms, 44 runs	
PCR-005 Protein A	Host cell protein	–	Log, 5 terms, 28 runs	PD-005-04, model selection
	Leached Protein A	Refitted (CA2)	Raw, no term, 28 runs	
	Fragments	Refitted (CA2)	Raw, 2 terms, 28 runs	
PCR-006 Low-pH viral inactivation	Aggregates	–	Raw, 2 terms, 18 runs	PD-006-04, model selection
	XMuLV clearance	Refitted (CA2)	Raw, 3 terms, 18 runs	
	Acidic variants	–	No adequate model, exact fit, 18 runs[a)]	
	Fragments	Refitted (CA2)	No adequate model, raw scale, 18 runs[b)]	
PCR-007 Cation exchange	Aggregates	–	Log, 8 terms, 28 runs	PD-007-01, model selection
	Host cell protein	–	Log, 6 terms, 28 runs	
	Fragments	Refitted (CA2)	No adequate model, raw scale, 28 runs[b)]	
PCR-008 Anion exchange	Host cell protein	Augmented (CA1)	Log, 2 terms, 40 runs	PD-008-01, augmentation PD-008-02, model selection
	XMuLV clearance	Refitted (CA2)	Raw, 4 terms, 28 runs	
	MVM clearance	Refitted (CA2)	Raw, 2 terms, 28 runs	
	Fragments	Refitted (CA2)	Raw, no term, 40 runs	
PCR-009 Virus filtration	MVM clearance	Refitted (CA2)	Raw, 1 term, 12 runs	PD-009-01, model selection
	XMuLV clearance	Refitted (CA2)	No adequate model, raw scale, 12 runs[b)]	
	Fragments	Refitted (CA2)	Raw, no term, 12 runs	

[a)] Fits exactly. The results have no scatter, so the full quadratic reproduces them and no scale is chosen. The model-adequacy criteria cannot be applied, and the output is judged on its measured results.
[b)] No adequate model on the scale shown. The output is judged on its measured results and sets no range.
Remedy: the remedy that the remedy rule of this investigation gave each full quadratic model that failed one of the three tests of PCMP-001 §7.2 that this investigation applies, on the planned runs, with Protein A response-surface run 23 at its design elution buffer pH. A dash means no remedy, because the re-analysis found no test failed. Model in the report: the model selected under corrective action CA3, with its scale, its terms besides the intercept and the block term, and the runs in which the output was measured.

Each report records the augmentation and the model selection as discrepancies from its plan, identified as plan discrepancies (PD). A discrepancy from the plan records work or an analysis that the approved plan did not foresee. A deviation records a departure from a procedure or a GMP requirement. No procedure and no GMP requirement was departed from, so neither is recorded as a deviation. The production bioreactor report (PCR-003) records the production bioreactor augmentation as PD-003-02, and the anion exchange report (PCR-008) records the anion exchange augmentation as PD-008-01. Each of the 6 reports with a response-surface design records the model selection (PD-003-03 in PCR-003, PD-005-04 in PCR-005, PD-006-

04 in PCR-006, PD-007-01 in PCR-007, PD-008-02 in PCR-008, PD-009-01 in PCR-009). The process characterization master plan (PCMP-001) and the step plans are not amended.

9.1 Acceptance criteria

Model selection was adopted on 2026-09-15, before the investigation into acceptance criteria set after the data were seen (INV-001) closed on 2026-09-18 and fixed the intermediate acceptance criteria. The verification of the platform worst cases behind the criteria in force therefore used the selected models of the planned runs (Section 8.4). In force, the verification widened 3 criteria by replacing a platform worst case with the one-sided 95 % prediction bound of a single A-Mab batch, the single-batch bound, for aggregates at the production bioreactor, host cell protein in the Protein A pool and host cell protein at cation exchange, and the criterion for aggregates at the production bioreactor was also tightened by another downstream step. The same verification on the full quadratic models, the analysis the plans prescribed, gives different values for 5 criteria. Against them, model selection made 4 criteria wider, by 4 %, 4 %, 7 % and 7 %, and kept 1 at its a priori value. Table 9.2 lists them, with the downstream worst cases whose values differ between the two verifications.

Table 9.2: Intermediate acceptance criteria that differ between the verification on the full quadratic models and on the selected models.

Criterion	Upper limit			Downstream worst case	Value in the derivation		
	A priori	Without model selection	In force		Platform	Full quadratic	Selected model
Production bioreactor, acidic variants (%)	39.0	39.6	39.0	Low-pH viral inactivation, acidic variant increase (percentage points)	1.00	0.400	1.00[a)]
Production bioreactor, aggregates (%)	9.30	6.19	6.46	Low-pH viral inactivation, aggregate increase (percentage points)	0.700	0.558	0.559
				Cation exchange, aggregate clearance (fold)	2.00	1.35	1.40
Protein A, host cell protein (ng/mg)	16,000	16,000	17,069	Anion exchange, host cell protein clearance (fold)	4.00	4.00[b)]	4.27
Low-pH viral inactivation, aggregates (%)	10.0	6.75	7.02	Cation exchange, aggregate clearance (fold)	2.00	1.35	1.40
Cation exchange, host cell protein (ng/mg)	400	400	427	Anion exchange, host cell protein clearance (fold)	4.00	4.00[b)]	4.27

[a)] The selected model of acidic variants at low-pH viral inactivation fits exactly and is not adequate, so its bound of 0.400 widens nothing and the platform value is used.
[b)] The full quadratic model of host cell protein at anion exchange is not adequate, because it fails the limit on overfitting. Its bound of 4.28 therefore widens nothing, and the platform value is used.
Upper limit: the intermediate acceptance criterion on the output of the step, a priori, after the verification on the full quadratic models (without model selection), and after the verification on the selected models (in force). The a priori limits start from the drug substance limits of 40 % acidic variants, 5 % aggregates and 100 ng/mg host cell protein. Value in the derivation: the value of the downstream worst case that the derivation uses on the full quadratic model and on the selected model. It is the platform value unless the verification replaced it by the one-sided 95 % prediction bound of a single A-Mab batch, at the least favourable combination of the NOR as characterized, from the planned runs. Limits and values are calculated before rounding.

The differences between the criteria from the two verifications come from changes at 3 downstream steps:

1. Anion exchange. The full quadratic model of host cell protein fails the limit on overfitting, so its single-batch bound of 4.28 cannot widen a criterion, and the platform worst-case clearance fold of 4 enters the derivation. The selected model is adequate, and its bound of 4.27 replaces the platform fold. The host cell protein criteria at cation exchange and in the Protein A pool therefore widen, because both lie upstream of anion exchange.
2. Cation exchange. The selected model of aggregate is on the log scale with 8 terms, and it gives a different prediction and a different single-batch bound from the full quadratic model. The predicted aggregate fold is worse than the platform fold on both models, so the bound replaces it on both. It tightens the aggregate criteria at low-pH viral inactivation and at the production bioreactor, which lie

upstream of cation exchange, by a different amount on each model. For host cell protein the predicted fold meets the platform fold of 40 on both models and its single-batch bound does not, so the platform fold stays on both, and no criterion moves because of host cell protein at cation exchange.

3. Low-pH viral inactivation. For aggregate, the single-batch bound of the increase is better than the platform value on both models, differs slightly between them, and enters the production bioreactor criterion (Section 8.4 gives why it is smaller than the platform value). For acidic variants, the full quadratic model reproduces every run exactly. The verification on the full quadratic models found that none of its tests failed and took its bound of 0.400 percentage points, which would have widened the upper limit of acidic variants at the production bioreactor to 39.6 %. Model selection does not accept an exact fit as adequate (Section 8.5.3). The platform value of 1.00 therefore stays, and the upper limit keeps its a priori value of 39.0 %.

Each widening rests on the one-sided 95 % prediction bound of a single batch at the least favourable combination of the NOR as characterized, from an adequate selected model. A check of a selected model against the three tests is optimistic (Section 8.1), so Table 9.3 shows what depends on the criteria that model selection moved. Judged against the criteria that the full quadratic models would have given, no output changes its conclusion over the NOR in force. One proven acceptable range moves. For host cell protein in the Protein A pool, the lower end of the elution buffer pH range moves from 3.21 to 3.24, and both lie below the NOR of 3.4 to 3.7.

Two qualifications of a scale-down model do depend on the wider criteria. A scale-down model is qualified for an output when its conclusion over the NOR is the same with the offset at either end of its 90 % confidence interval (PCMP-001 §7.1). Against the criteria without model selection, the scale-down model would not be qualified for host cell protein in the Protein A pool or at cation exchange. In both cases the low end of the offset interval decides. In the Protein A pool, with the offset at the low end of its interval, a ratio of 0.89, the upper limit of the two-sided 95 % prediction interval of a single batch over the NOR, at manufacturing scale, is 16,763 ng/mg. That is above the criterion of 16,000 ng/mg that the full quadratic models give, and below the criterion in force of 17,069 ng/mg. At cation exchange, with the offset at 0.63, the same limit is 414 ng/mg, against 400 ng/mg without model selection and 427 ng/mg in force. These two qualifications rest on the criteria that model selection widened.

The wider criteria concern host cell protein and aggregates, which are attributes of the drug substance specification, with limits of 5 % aggregates and 100 ng/mg host cell protein. A step output at its criterion still gives a drug substance within its limit, as long as each downstream step clears at least the value that the derivation uses. For host cell protein these are a fold of 40 at cation exchange, the platform worst case, and 4.27 at anion exchange, its single-batch bound. For aggregates they are a fold of 1.40 at cation exchange and an increase of 0.559 percentage points at low-pH viral inactivation, both single-batch bounds. The integrated process model simulates batches through every step of the train in sequence, from the step models with their run-to-run variation. The master plan states that each process parameter of each batch is drawn from a normal distribution centred on its set-point, with a standard deviation of one sixth of the width of its NOR (PCMP-001 §6.4 and §6.5).

None of the 2,000 batches that the integrated process model simulated gives an out-of-specification drug substance for either attribute. With no failure in 2,000 batches, the one-sided 95 % upper bound on the rate of failure is 0.15 %.

Table 9.3: What depends on the criteria that model selection moved.

Criterion	In force	Without model selection	Conclusion over the NOR	Proven acceptable ranges	Scale-down model qualified
Production bioreactor, acidic variants (%)	18.0–39.0	18.0–39.6	not met / not met	Same	yes / yes
Production bioreactor, aggregates (%)	≤ 6.46	≤ 6.19	met / met	Same	yes / yes
Protein A, host cell protein (ng/mg)	≤ 17,069	≤ 16,000	met / met	Elution buffer pH, lower end 3.21 / 3.24	yes / no
Low-pH viral inactivation, aggregates (%)	≤ 7.02	≤ 6.75	met / met	Same	yes / yes
Cation exchange, host cell protein (ng/mg)	≤ 427	≤ 400	met / met	Same	yes / no

In force and Without model selection: the criterion in force and the criterion that the verification on the full quadratic models gives. Each pair in the other columns gives the result against the criterion in force, then against the criterion without model selection. The conclusion over the NOR is judged at the NOR in force. Proven acceptable ranges: Same where every range of the output is unchanged.

The augmentation runs moved no criterion. Both augmentations ended after INV-001 closed, on 2026-09-24, and the verification uses only the planned runs of each step. Repeated on all runs of anion exchange, the verification gives the same bound for host cell protein, 4.27, so the augmentation runs would not have moved the criteria either.

9.2 Conclusions over the NOR

The conclusion over the NOR of an output, whether it meets its criterion at every combination of NOR low, set-point and NOR high, does not depend on the choice between the full quadratic model of the planned runs and the model that the report uses. For every output with a criterion the two give the same conclusion, at the NOR as characterized and at the NOR in force. Two of these conclusions are not met on either model, and both rested on full quadratic models that fail the limit on overfitting:

1. XMuLV clearance at low-pH viral inactivation is judged on the mean LRF at the least favourable combination of NOR low, set-point and NOR high, because viral clearance is validated at worst-case conditions (PCMP-001 §6.4). The criterion is the platform claim of the step, 6.6 log10. The claims of the three viral clearance

steps add up to 16.72 log10, against a cumulative requirement of 16.7 log10. The selected model has the main effects of inactivation pH, hold time and temperature only. On it, the lowest mean LRF within the NOR as characterized is 6.16, at inactivation pH 3.6, hold time 60 min and temperature 19 °C, so the step claim is not met. The full quadratic model gives a lowest mean of 6.43 and the same conclusion. The low-pH viral inactivation report (PCR-006) moves the NOR of inactivation pH, hold time and temperature, and it states the conclusion over the NOR in force.

2. Acidic variants at the production bioreactor are judged on the 95 % prediction interval of a single batch at every combination of NOR low, set-point and NOR high. The lower limit of the criterion is 18 %, the lower limit of the drug substance specification carried to the step unchanged. In the selected model acidic variants fall as culture pH rises, as in the model of the A-Mab case study (CMC Biotech Working Group 2009, Table 3.16). The least favourable combination therefore lies at the NOR high of culture pH. On the selected model of all runs, the lower prediction limit at culture pH 6.95, culture temperature 34.5 °C, culture duration 18 days and CO2 partial pressure (pCO2) 130 mmHg is 16.2 % at manufacturing scale, so the criterion is not met. The predicted median at the same settings is 18.4 %, within the criterion. On the full quadratic model of the planned runs the lower prediction limit at the same settings is 14.7 %, and the criterion is not met either. The production bioreactor report (PCR-003) states how this result is handled.

Acidic variants at low-pH viral inactivation, fragments at low-pH viral inactivation, fragments at cation exchange and XMuLV clearance at virus filtration have no adequate model and are judged on their measured results (Table 8.4).

9.3 Other documents

The other documents of the package are affected as follows:

1. INV-001, the investigation into acceptance criteria set after the data were seen, is not reopened. Its comparison of the proven acceptable ranges of Version 1.0 and Version 2.0 uses the models that Version 1.0 fitted, so that it shows what the change of criteria moved. The intermediate acceptance criteria in force rest on the selected models (Section 9.1).
2. The process characterization master plan (PCMP-001 §7.2) allows a failing model to be refitted, or not used to set a range. The augmentation is the rule of this investigation. The master plan is not changed for model selection, which each report records as a discrepancy from its plan.
3. This investigation requires the process characterization master report (PCMR-001) Version 2.0 to consolidate the conclusions of the step reports as they stand after the remedies and the model selection.

The investigation has no product impact. No batch was dispositioned on these models, and the Version 1.0 documents were on hold for BLA use.

10 Corrective and preventive actions

Table 10.1 lists the corrective and preventive actions. Model selection, corrective action CA3, replaced in the reports the refits by backward elimination of corrective action CA2. Corrective action CA1, the augmentation of the designs that the remedy rule names, has two parts. The augmentation runs of CA1 were complete at closure. The second part of CA1, the re-evaluation of the augmented designs on all runs, is carried by the Version 2.0 reports of the two steps, due on 2026-09-30, together with the models chosen by model selection. Both were open at closure.

Preventive action PA1, the sample-size calculation for each new characterization plan, checks each new design by simulation before its plan is approved, at the expected effects and with the analysis the plan states. The simulation of Section 6 used the full quadratic model, the analysis the Version 1.0 plans prescribed. Preventive action PA2 revises SOP-1002, Design, Execution and Statistical Analysis of DoE Studies, so that a plan states its model-selection procedure.

Table 10.1: Corrective and preventive actions of INV-002.

ID	Action	Owner	Due	Status at closure
CA1	Augment each response-surface design that the rule names, with the number of runs INV-002 calculates, and re-evaluate its models on all runs with a block term	MSAT	2026-09-30	runs complete at closure; the re-evaluation on all runs is carried by the Version 2.0 reports, open at closure
CA2	Refit every other failing model by hierarchical backward elimination, and judge a response with no adequate model on its measured results	MSAT	2026-09-30	complete at closure
CA3	Analyse every response-surface output by model selection (Box-Cox scale, AICc over all hierarchical submodels of the full quadratic, block kept) and judge the selected model against PCMP-001 §7.2; each report records it as a discrepancy from its plan	MSAT	2026-09-30	adopted on 2026-09-15; applied by the Version 2.0 reports, open at closure
PA1	Check each new characterization plan before approval, by simulation at the expected effects and with the analysis the plan states, for a probability of success of at least 0.90 that its planned models meet the lack-of-fit test and the limit on overfitting	Statistics	2026-12-31	open at closure
PA2	Revise SOP-1002 so that a characterization plan states its model-selection procedure	Quality Assurance	2026-12-31	open at closure

11 Effectiveness check

The actions are checked for effectiveness by 2027-06-30. The check is recorded in three parts:

1. For the next two characterization plans, the sample-size calculation of preventive action PA1 documented for every planned model before approval.
2. At their first analysis, for every attribute with a criterion, a selected model that meets the first three model-adequacy criteria of PCMP-001 §7.2 (the model F-test, the lack-of-fit test and the limit on overfitting) and whose 95 % prediction intervals contain the step's confirmation runs at least as often as a one-sided binomial test at the 0.05 level allows.
3. If no plan is approved by the due date, the check is made on the next characterization study of A-Mab.

The second part asks for the three tests that this investigation applies, so a selected model must be adequate as Section 3 defines it. The second part also tests the prediction of new runs. Under a correct model each confirmation run falls inside its 95 % prediction interval with a probability of 0.95, and the number inside follows a binomial distribution. The test rejects the model when that number is so low that a correct model would give it with a probability below 0.05. With 16 confirmation runs at least 14 must fall inside, and with 12 at least 10. Applied to the augmentation runs of this investigation, the binomial test rejects the models of high mannose and aggregates at the production bioreactor that were selected on the planned runs (Section 8.5.1).

12 Conclusion and closure

The investigation confirmed the classification as minor, on the grounds recorded when it was opened (no GMP batch was made or dispositioned on the analysis, no product quality impact, and the finding concerns the statistical analysis of development data, not their integrity). No range was set from a failing model at Version 2.0.

The root cause is that the plans prescribed the full quadratic model as the predictive model, with no model selection, on designs that are small for it. The Version 1.0 plans set no numeric limit on how far predicted R^2 may fall below R^2, against which the size of a design could have been checked. A remedy was assigned to every failing model by a rule approved before any new run, and both augmentations were run by 2026-09-24. Model selection then replaced the refits for every output. Once selected, the models that the augmentations were made for meet the three tests of the model-adequacy criteria (a significant model F-test ($\alpha = 0.05$), no significant lack of fit ($\alpha = 0.05$) and R^2 minus predicted R^2 of at most 0.20) on the planned runs alone. On all runs the full quadratic model with a block term, which the augmentations were sized for, meets the lack-of-fit test and the limit on overfitting for acidic variants and host cell protein, and not for high mannose. The models of high mannose and aggregates at the production bioreactor, selected on the planned runs, predicted fewer of the new runs within their intervals than a binomial test allows, and the reports use the models of all runs. After selection 4 outputs have no adequate model, and every measured result of these outputs meets its criterion.

The intermediate acceptance criteria in force were fixed on the selected models at the closure of the investigation into acceptance criteria set after the data were seen (INV-001). Against the criteria that the full quadratic models give, model selection made 4 intermediate acceptance criteria wider and kept the upper limit of acidic variants at the production bioreactor at its a priori value. No conclusion over the NOR depends on that difference, and one proven acceptable range moves at its lower end, outside the NOR. The qualification of the scale-down model for host cell protein in the Protein A pool and at cation exchange does depend on it. The augmentation runs moved no criterion.

At closure the augmentation runs of corrective action CA1 and the refits of corrective action CA2 were complete. The re-evaluation on all runs of CA1 and the application of model selection, corrective action CA3, in the Version 2.0 reports were open, due on 2026-09-30. The preventive actions were open. INV-002 was closed on 2026-09-28.

References

Box, George E. P., and David R. Cox. 1964. "An Analysis of Transformations." *Journal of the Royal Statistical Society, Series B (Methodological)* 26 (2): 211–52.

Burnham, Kenneth P., and David R. Anderson. 2002. *Model Selection and Multimodel Inference: A Practical Information-Theoretic Approach*. 2nd ed. New York: Springer.

CMC Biotech Working Group. 2009. "A-Mab: A Case Study in Bioprocess Development." CASSS.

Fedorov, Valerii V. 1972. *Theory of Optimal Experiments*. New York: Academic Press.

Hurvich, Clifford M., and Chih-Ling Tsai. 1989. "Regression and Time Series Model Selection in Small Samples." *Biometrika* 76 (2): 297–307.

International Council for Harmonisation. 2023. "ICH Q5A(R2): Viral Safety Evaluation of Biotechnology Products Derived from Cell Lines of Human or Animal Origin." ICH.

Montgomery, Douglas C. 2017. *Design and Analysis of Experiments*. 9th ed. Hoboken, NJ: Wiley.